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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SDS RISER WITH VT H07																	
				Date Prep. 15.11.2018		PROJECT TITLE				COVER SHEET		Scale		= H07			
				Prepared YR		MV Switchgear for MNA Serang Plant				MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Customer		PT Multimas Nabati Asahan & 310			
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F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet	Scale		= H07
					Prepared	YR	MV Switchgear for MNA Serang Plant					MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Customer	PT Multimas Nabati Asahan
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION	-CA	CAPACITORS
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION	-CA1	CAPACITOR FOR FRONT VENTILATOR
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)	-CA2	CAPACITOR FOR REAR VENTILATOR
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION	-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.
B	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION	-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION	-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION	-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT
	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION	-EB1	HEATER LOCATED IN CABLE COMPARTMENT
C	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION	-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT
	-BGK	POSITION SWITCH OF KEY LOCK	-EB3	HEATER LOCATED IN MOTOR CABINET
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1	-EB4	HEATER FOR VENTILATION TEST
	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS	-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.
D	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION	-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION	-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION	-FA1	SURGE ARRESTER LOCATED ON PHASE L1
	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION	-FA2	SURGE ARRESTER LOCATED ON PHASE L2
E	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION	-FA3	SURGE ARRESTER LOCATED ON PHASE L3
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION	-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION	-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1
	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION	-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2
F	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION	-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION	-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION	-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR	-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1
	-BGV2	POSITION SWITCH OF REAR VENTILATOR	-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2
	-BM	HYGROSTAT	-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING	-FCT	THERMAL OVERLOAD RELEASES
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING	-GA	GENERATORS
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING	-GQ1	FRONT VENTILATOR
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING	-GQ2	REAR VENTILATOR
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER	-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2	-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3	-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE
	-BR	FLAME DETECTORS, SMOKE DETECTORS	-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS
	-BT	THERMOSTAT	-KFC	CLOSING RELAYS OR CONTACTORS
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS	-KFI	INTEGRATED CIRCUITS
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS	-KFL	LOCKOUT RELAY
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM	-KFM	MICROPROCESSORS
	-BU	BUCHHOLZ RELAY	-KFN	ANTIPUMPING RELAYS
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING	-KFO	OPENING RELAYS OR CONTACTORS
	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING	-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)
	-BZ,-BZ4	MULTIFUNCTION SENSORS	-KFR	RECLOSING RELAYS
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1	-KFS	SYNCHRONIZING DEVICES
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2	-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3	-KFZ	CONTROLLERS
			-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER
			-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD
			-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1
			-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2
			-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS

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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
B	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
C	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
D	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
E	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
F	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
G	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
H	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
I	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
J	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
K	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
L	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
M	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
N	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
O	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
P	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
Q	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
R	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
S	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
T	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
U	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
V	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
W	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
	-QBD4	TWO-WAY DISCONNECTOR		
X	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

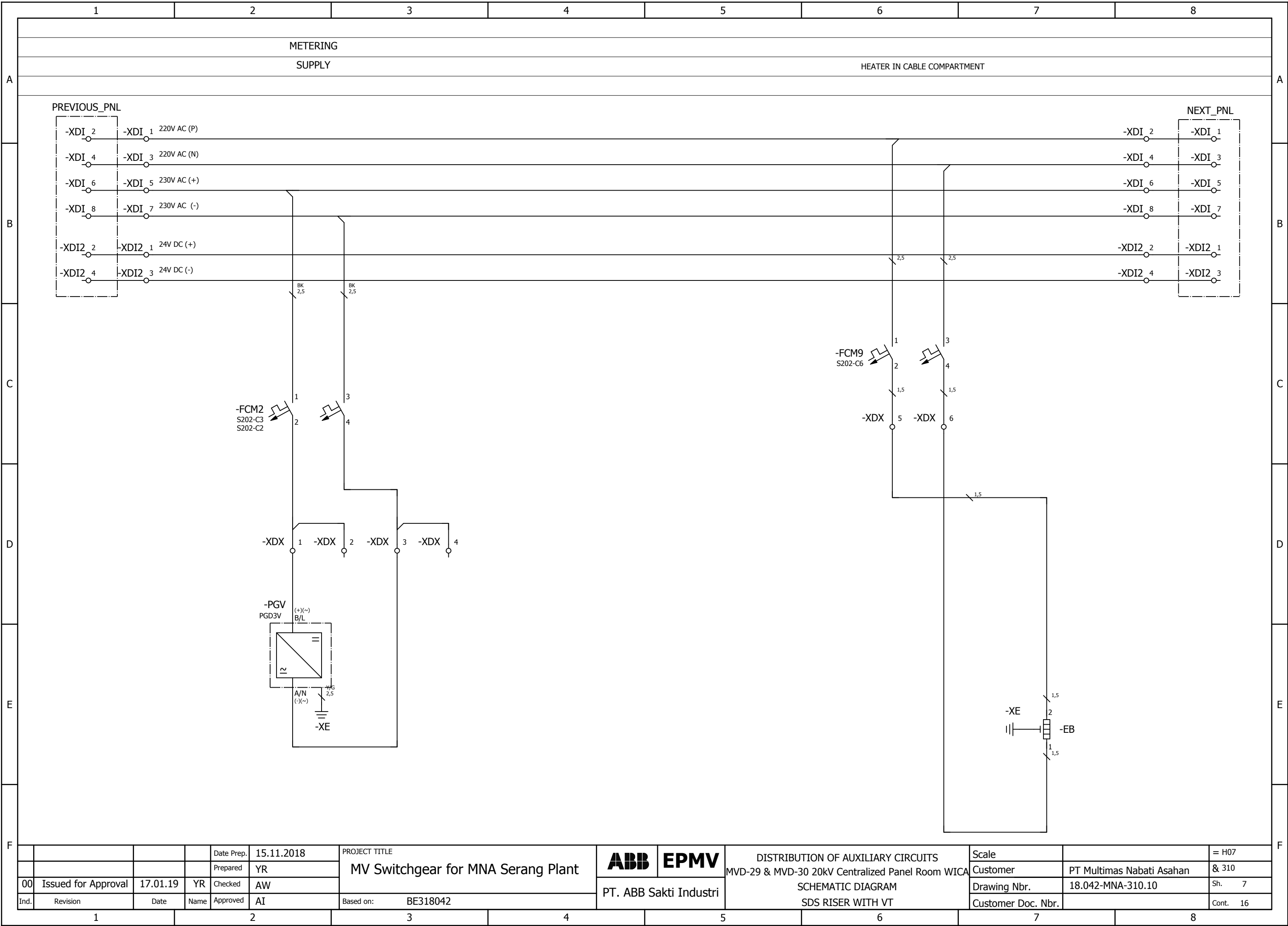
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A	-TFA		ACTIVE POWER TRANSDUCERS		-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS		A							
	-TFC		CURRENT TRANSDUCERS			-XD12			TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						
	-TFF		FREQUENCY TRANSDUCERS		-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION									
	-TFJ		ACTIVE ENERGY TRANSDUCERS			-XD14			TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS						
	-TFK		REACTIVE ENERGY TRANSDUCERS		-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS									
	-TFM		MULTIFUNCTION TRANSDUCERS			-XD17			TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS						
	-TFP		POWER-FACTOR TRANSDUCERS		-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS									
	-TFQ		REACTIVE POWER TRANSDUCERS			-XD19			TERMINAL BLOCK FOR SPECIAL USE						
	-TFS		SIGNAL CONVERTERS		-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD									
	-TFV		VOLTAGE TRANSDUCERS			-XDJ1			TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS						
B	-WA		BUSBARS		-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT		B							
	-WBC		POWER CABLES			-XDS			CURRENT SOCKETS						
	-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.		-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB									
	-WGC		CONTROL CABLES			-XDU			CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK						
	-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)		-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK									
	-WH		OPTICAL FIBRES			-XDV			TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS						
	-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS		-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1									
	-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			-XDV2			CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2						
	-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2		-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3									
	-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			-XDV4			CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR						
	-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER		-XDX	SUPPORT TERMINAL BLOCK									
	-XDB		CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS			-XE1			EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE						
	-XDB1		CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)		-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE									
	-XDB2		CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER			-XE5			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE						
	-XDB10...-XDB89		CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS		-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE									
	-XDB7		CONNECTOR FOR TRUCK POSITION CONTACTS			-XE7			EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE						
	-XDB91		CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE									
	-XDB92		CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			-XE9			EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT						
	-XDB93		CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR												
	-XDB95		CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET												
D	-XDC		CUSTOMER TERMINAL BLOCK												
	-XDC1		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER												
	-XDC2		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR												
	-XDC3		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT												
	-XDC4		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS												
	-XDC5		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER												
	-XDE		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE												
	-XDE1		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE												
	-XDE2		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2												
	-XDE3		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2												
	-XDF		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS												
	-XDF1		CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR												
	-XDF2		CONNECTOR FOR REAR VENTILATOR CIRCUITS												
	-XDF3		CONNECTOR FOR FRONT VENTILATOR CIRCUITS												
	-XDH		TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)												
	-XDI		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)												
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DISTRIBUTION OF AUXILIARY CIRCUITS MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale		= H07
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 310
00	Issued for Approval	17.01.19	YR	Checked	AW		PT. ABB Sakti Industri	SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-310.10	Sh.
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	SDS RISER WITH VT			Customer Doc. Nbr.		Cont. 16

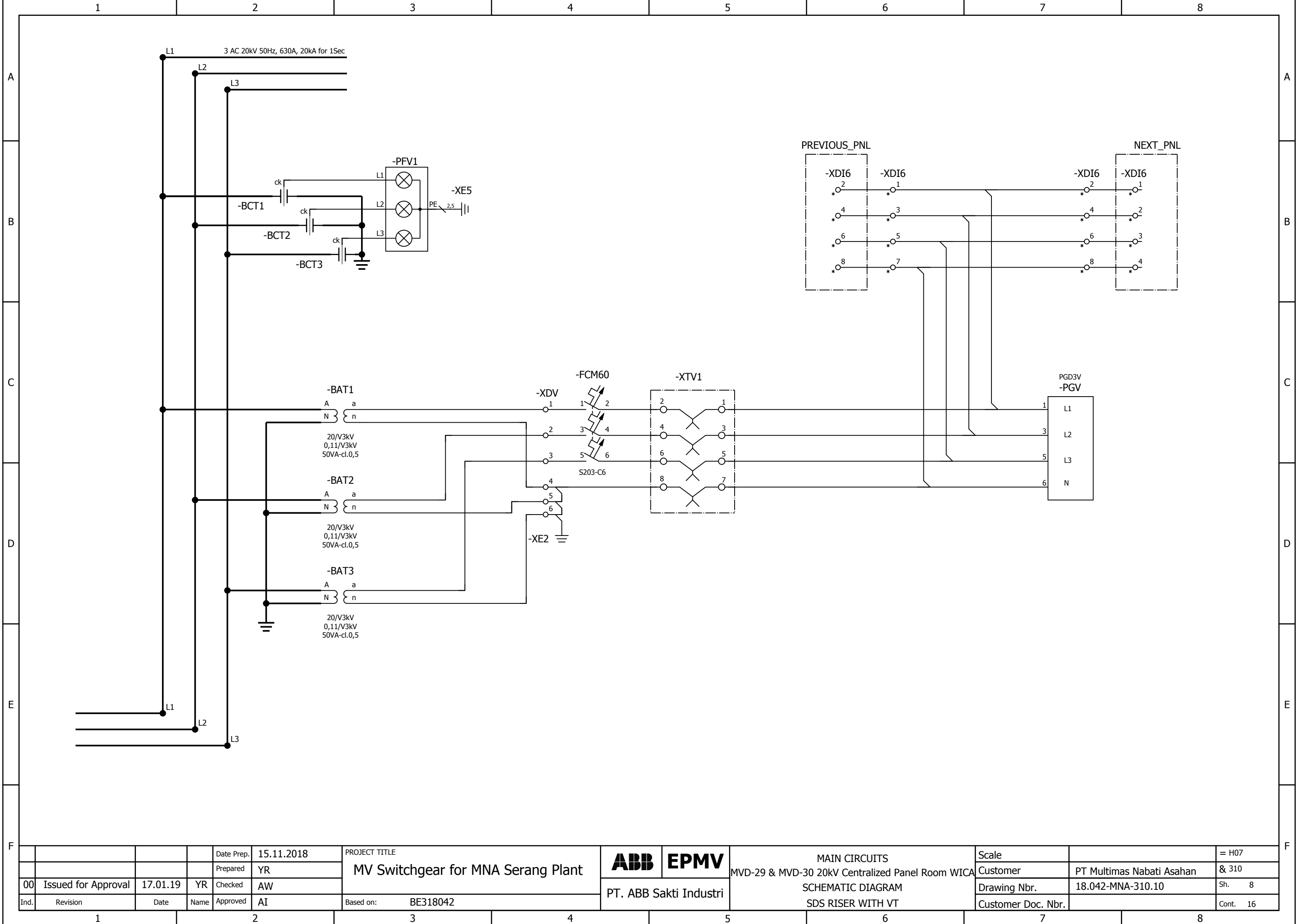


Diagram illustrating the Main Circuits Schematic Diagram for the MV Switchgear for MNA Serang Plant, showing the connection between the incoming supply (L1, L2, L3) and the outgoing distribution (L1, L2, L3, N).

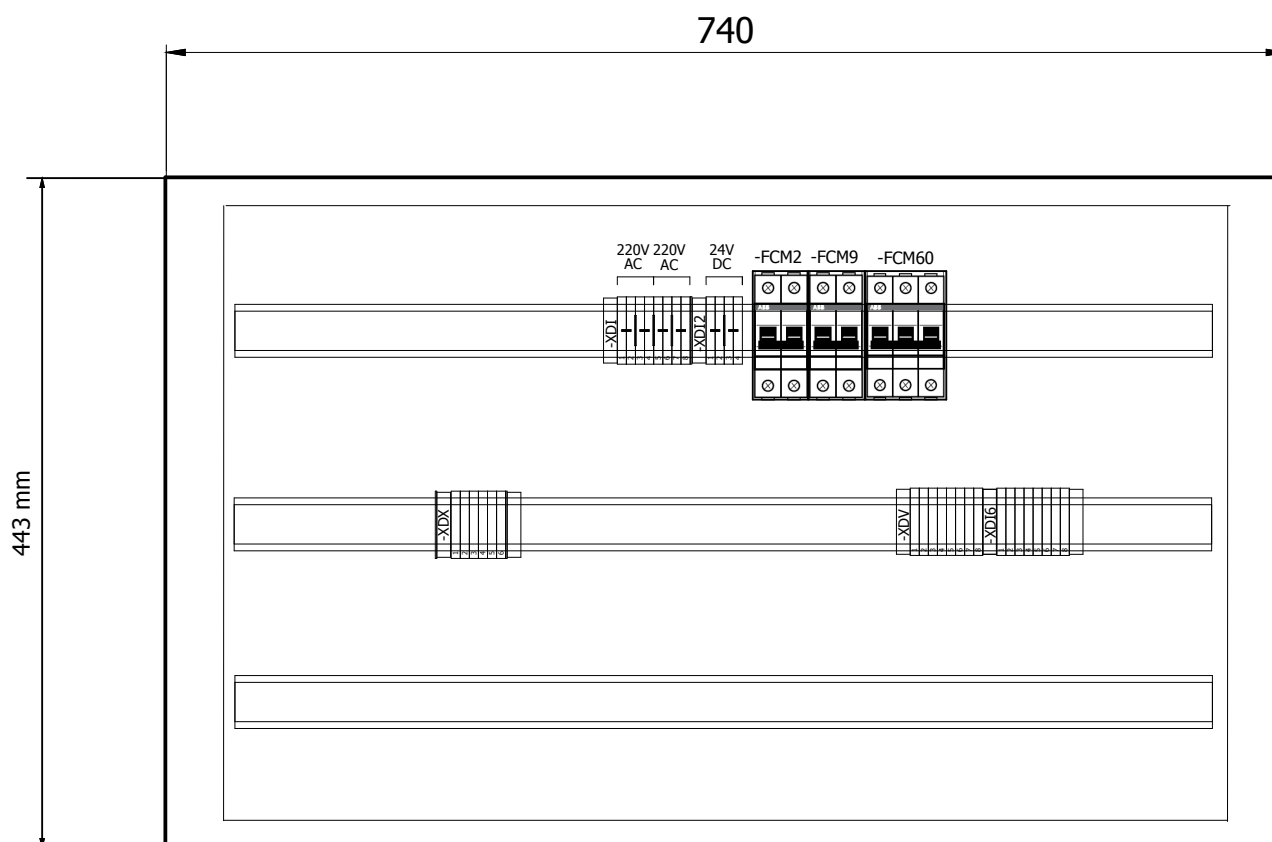
The diagram includes the following components and connections:

- Incoming Supply:** L1, L2, L3 (3 AC 20kV 50Hz, 630A, 20kA for 1Sec).
- Protection and Control:** -BCT1, -BCT2, -BCT3 (Circuit Breakers); -PFV1 (Protective Function Unit); -XE5 (Earth Fault Protection).
- Transformers:** -BAT1, -BAT2, -BAT3 (20/V3kV, 0,11/V3kV, 50VA-cl.0,5).
- Switchgear:** -XDV (Disconnector); -FCM60 (Circuit Breaker); -XTV1 (Transfer Switch).
- Earth Fault Protection:** -XE2 (Earth Fault Protection).
- Outgoing Distribution:** L1, L2, L3, N (PGD3V -PGV).
- Interlocks:** -XDI6 (Interlocking Device) connecting to PREVIOUS_PNL and NEXT_PNL.

The diagram shows the flow of power from the incoming supply through the switchgear components to the outgoing distribution busbars. The interlocking device (-XDI6) ensures safe operation by coordinating with the previous and next panels.

Rev	Description	Date	By	Check	Appr	Based on:	Project Title	Scale	Customer	Customer Doc. Nbr.
00	Issued for Approval	17.01.19	YR	AW	AI	BE318042	MV Switchgear for MNA Serang Plant	MAIN CIRCUITS MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SDS RISER WITH VT	PT Multimas Nabati Asahan	18.042-MNA-310.10

[illegible]

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	 	LV BOX LAYOUT MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SDS RISER WITH VT	Scale		= H07
				Prepared	YR				Customer	PT Multimas Nabati Asahan	& 310
00	Issued for Approval	17.01.19	YR	Checked	AW				Drawing Nbr.	18.042-MNA-310.10	Sh. 10
Ind.	Revision		Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 16

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Summarized parts list															
IDABB_F02_005-IDABB															
Carry over: 0.00															
TAG ID		DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS			
-EB		HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.7:E			
-FCM2		MCB S202-C2 MCB. 2P, 2A				ABB		2CDS252001R0024		1		/7.2:C			
-FCM9		MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		1		/7.6:C			
-FCM60		MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/8.4:C			
-PGV		Digital AC Voltmeter Model : DPM PGD3V, 48 X 96 MM				RISHABH		DPM PGD3V		1		/7.2:D			
-XTV1		Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSPTT-04C		1		/8.5:C			

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